

## ARCESIUM INTERNSHIP INTERVIEW EXPERIENCE 2021 – 22

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**Arcesium** is a global financial technology and professional services firm, delivering solutions to some of the world's most sophisticated financial institutions, including hedge funds, banks, and institutional asset managers.

There were 4 rounds of selection process, 1 online test (23/08/2021) and 3 technical rounds (27/08/2021).

### **Online Test :** (80 mins)

Platform : HackerRank

### **Section 1 :** Aptitude – 15 MCQs (20 mins) :

Questions difficulty level ranged from medium to high. This section had negative marking.

### **Section 2 :** Technical MCQs – 15 MCQs (15 mins) :

Questions were challenging and covered a variety of concepts. This section also had negative marking. Concepts covered :

- Predict the output for C++ and java code snippets – based on pointer manipulation in C++, inheritance based questions, interfaces in java.
- DBMS – SQL queries
- OS – process, memory management, etc.

### **Section 3 :** Coding – 2 questions (45 mins)

1. **Naruto Clan Curse** – It is an application of level-order traversal on binary tree, where we have to return the avg of all the unaffected node values from each level. Min value of the extreme nodes of each level is calculated. A node is affected when its value is less than the above min value.
2. **ATM Change problem** – It is a dynamic programming problem. A req amount, preferred denomination and set of available denominations are given. We have to return the number of ways to make change.

Ex : Amount = 1200 Preferred denomination = 500 denominations = [ 100, 200, 500]

1.  $1200 = 500 + 500 + 100 + 100$

2.  $1200 = 500 + 500 + 200$

Output : 2

Solution : Modification of <https://www.geeksforgeeks.org/coin-change-dp-7/>

### **Technical Round 1 :** (1 hour)

This round was a **pair coding round**.

I was given a **backtracking** question. First asked to write the pseudo code and optimize it. At last, code it in the language of your choice.

Question : Given  $n \times n$  matrix filled with 1, 2, 3 . Find the path with max reward from src to dest.

1 -> move one cell right

2 -> move one cell down

3 -> move one step either right or down

1 2 1 2

3 3 1 2

1 2 1 2

3 1 1 2

Src = (0, 0)

Dest = (3, 3)

Max reward path = 1 -> 2 -> 3 -> 1 -> 2 -> 2 -> 2 = 13

## Technical Round 2 : ( 1 hour)

This round assessed the **technical skills and projects** mentioned in the resume.

1. Asked to explain the database design of the **DBMS project** mentioned in my resume.  
Discussed about the scalability and potential security issues of the project.
2. Several concepts of DBMS such as – Indexing, SQL queries, views – were also discussed as a part of project discussion.
3. Brief discussion on several **OOPs concepts** and **Java**, as I mentioned Java in my resume, on :
  - ✓ JVM and platform independency of java
  - ✓ Garbage collection in java
  - ✓ Exception handling
  - ✓ Finalize keyword
  - ✓ Virtual functions
  - ✓ Method overloading vs over-riding
  - ✓ Inheritance
  - ✓ Abstract , static in java
4. Questions from the following **OS** Concepts were asked :
  - ✓ Interrupts
  - ✓ Paging & Page fault
  - ✓ Thrashing
  - ✓ Deadlocks and its avoidance ( Banker's algorithm)

### Technical Round 3 : (1 hour)

I was given few **coding questions** and come up with an algorithm with specified time and space complexities.

**Note** : Discuss all possible edge cases and efficiency of the algorithm in each case.

1. Given an array of n consecutive integers where one of the element is over-written by another element in the array. Now this array is shuffled and find the element being over-written with minimum time as possible without any auxiliary space.

Ex:  $A[] = [51, 52, 53, 54, 55] \rightarrow [51, 52, 53, 52, 55] \rightarrow [52, 55, 51, 53, 52]$

Result = 54 , as 52 over-writes 54

2. Given a sorted array where all elements except one element is repeated twice. Find the distinct element with minimum time and  $O(1)$  space.

Ex :  $A[] = [1, 1, 2, 2, 3, 4, 4, 5, 5]$  Result = 3

3. Given an mathematical expression place the parenthesis such that value of the expression is maximum.

Ex :  $5*2-4/1+3$

Ref : <https://www.geeksforgeeks.org/all-ways-to-add-parenthesis-for-evaluation/>

### Key takeaway and concepts to be covered :

1. Prepare “Tell about yourself” as attractive and concise, covering the key points of your resume. Answer confidently and try the questions you don’t know, don’t leave them unanswered. They’ll test your knowledge on the skills you mentioned in the resume, so be clear on those concepts.
2. When given a coding question, first explain your intuition and analyze it’s time and space complexities and try to optimize it as much as you can. Feel free to use a pen and paper to solve and also try to approach the problem mathematically if possible.
3. DSA – Arrays, Trees, Sorting, Heaps, Backtracking and Dynamic programming – among others.
4. DBMS – Database design and relation schema, Indexing and hashing, SQL, Views, Triggers.
5. OS – Basics of OS, Process and synchronization, deadlocks, memory management (especially Paging)
6. OOPS – Fundamentals, Inheritance types, Polymorphism types, Singleton class implementation (not asked to me, but worth learning).
7. Revise some common interview questions on the above mentioned concepts.
8. Company archives in GFG gives a prior idea on the online test and technical interviews. Also practice the coding questions asked in the past from GFG Practice.

Regards,

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